

Kevin Wei

kevwei@seas.upenn.edu | (505)-550-2166 | www.linkedin.com/in/kevwei | https://kevwei.com/

EDUCATION

University of Pennsylvania, School of Engineering and Applied Science, Philadelphia, PA May 2028

Bachelor of Science in Engineering in Computer Science & Computer Graphics (GPA: 3.80/4.00)

- *Relevant Coursework:* Discrete Math; Automata Theory; Program Design; Algorithms & Data Structures; Computer Systems; Computer Graphics; Computational Linear Algebra; 3D Figure Modeling; 3D Animation
- *Activities:* UPGRADE; Penn Spark; Penn Lion Dancing; Penn SIGGRAPH; Content Manager @ Penn Review

EXPERIENCE

Co-President | *UPenn Games Research and Development Environment (UPGRADE) Club* May 2025 – Present

- Organized events, meetings, a Halloween Game Jam, and led small groups to create **13 unique minigames**
- Hosted **AI & Gaming guest talk** with CTO and COO of **Medal.tv**, the world's largest gaming clipping tool
- **Produced workshops** on game dev basics, level design, modeling, shaders, polishing, UI/UX, and more
- Made dev tools, VFX, and camera code for Catanks, a club-wide Wii Tanks-esque game **published on Steam**

Founder & President | *Maria Carrillo High School (MCHS) Game Dev Club* September 2021 – June 2024

- **Founded new club** to teach C#, Unity, game design, pixel art, and music production to my local community
- Organized **guest talk by professional VFX artist** from Hidden Leaf Games to educate about games industry

PROJECTS

Space Minecraft | *C++, OpenGL, GLSL, Qt Creator* November 2025 – December 2025

- Minecraft but in space! Place blocks on Saturn's rings, walk around the Earth, explore the burning Sun's surface!
- Developed **deferred rendering engine from scratch** for effects such as screen-space ambient occlusion
- Translated flat voxel terrain into **smooth cubespheres**; created gravity, orbits, player-planet **velocity matching**
- Simulated **realistic atmospheres and sunrises/sunsets** using physically accurate Rayleigh and Mie scattering
- Created **LOD system** applying quadtree algorithm onto planetary surfaces, leading to **40x** performance increase

Physically Accurate Fluid Renderer | *C#, HLSL, Unity* June 2025 – August 2025

- Simulated Newtonian fluids using **Smoothed Particle Hydrodynamics** and Navier-Stokes equations
- Optimized runtime with bitonic merge sort and **3D spatial hashing** to support **260,000+ particles at 200 FPS**
- Real-time raymarching with **realistic refraction/reflection/Fresnel** equations for accurate light ray simulation

Flocks of Fish | *C#, Shader Graph, VFX Graph, HLSL, Unity* May 2025 – June 2025

- Hop in a toon-shaded submarine! Explore deep ocean of flocking fish, sharks, whales, coral, bubbles, god rays
- Developed **GPU-accelerated** Boids algorithm and achieved stable performance of **20,000+ fish at 200 FPS**

Flying Through Clouds | *C#, HLSL, Unity* April 2025 – May 2025

- Soar through **physically realistic clouds**, with raymarching to simulate forward scattering and light attenuation
- Built GPU-accelerated **Worley Noise** and **fractal Brownian Motion** tool to generate custom 3D cloud shapes

A Bear Game | *C#, Shader Graph, VFX Graph, Unity, Blender* December 2024 – April 2025

- Play as a bear who takes pictures of nature, islands, volcanoes, birds, fish, cherry blossoms, and skyscrapers
- Explore the **open world**: beaches, cities, forests. Drive on roads, talk to people, catch fireflies, and buy donuts
- Built **node-based dialogue** and **Bézier spline path editor** for easy, modular creation of **complex NPC behavior**

Descent | *C#, Unity, Team of 4* August 2024 – November 2024

- 3D snowboarding game with **endless procedural terrain**, mid-air flips/tricks system, and sand/snow biomes
- Composed **original 6-minute soundtrack** on MuseScore with 2 cellos, piano, harp, acoustic bass, and drum set

TECHNICAL SKILLS

Programming Languages: C#, C++, C, Java, Python, HLSL, GLSL, JavaScript, TypeScript, HTML, CSS, OCaml

Platforms/Tools: Unity Engine, Unreal Engine, UE Blueprints, Shader Graph, Visual Studio, Blender, ZBrush, Maya, DaVinci Resolve, After Effects, Aseprite, Audacity, Figma, React, TailwindCSS, Django, PostgreSQL, Git, NextJS, OpenGL